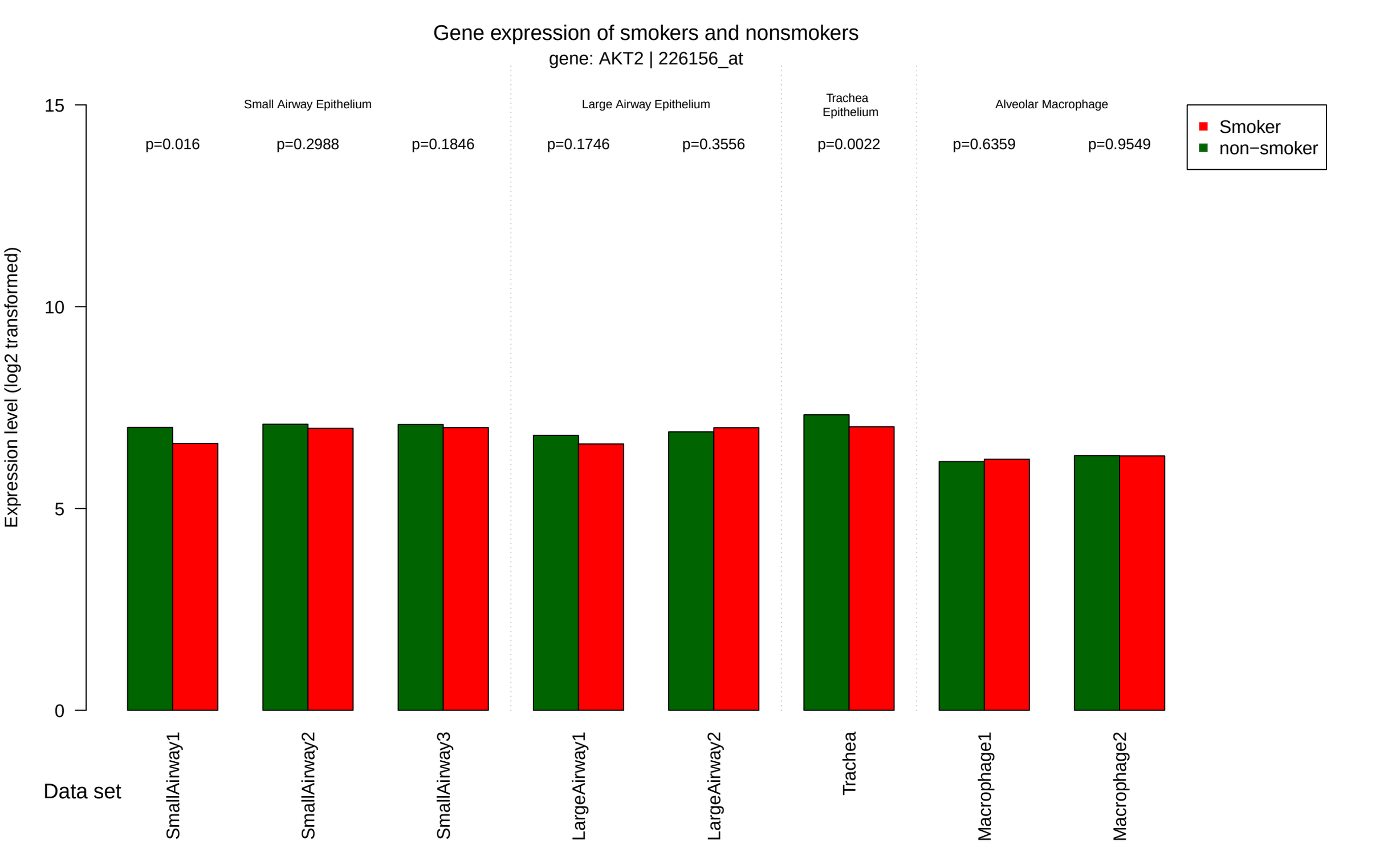
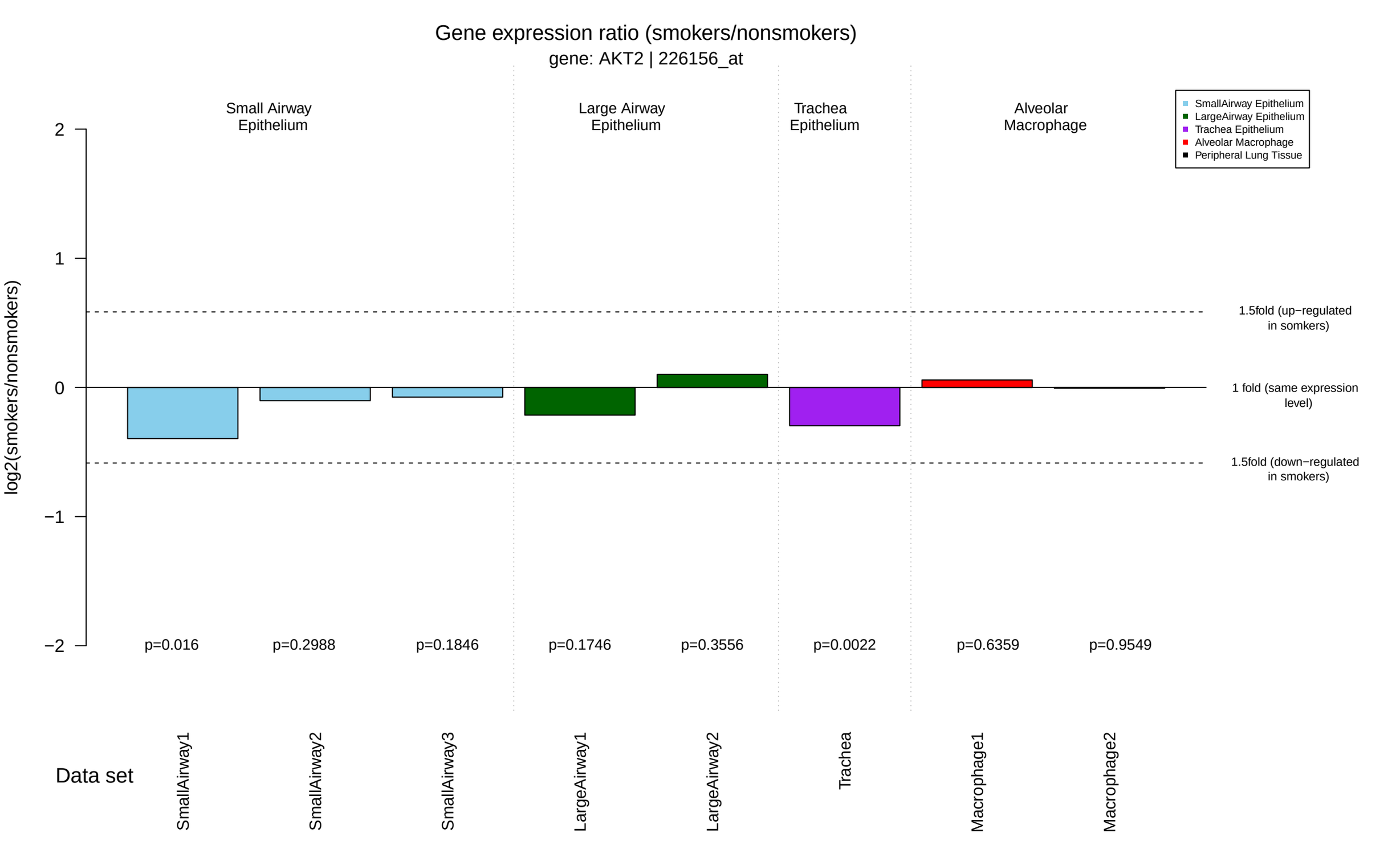
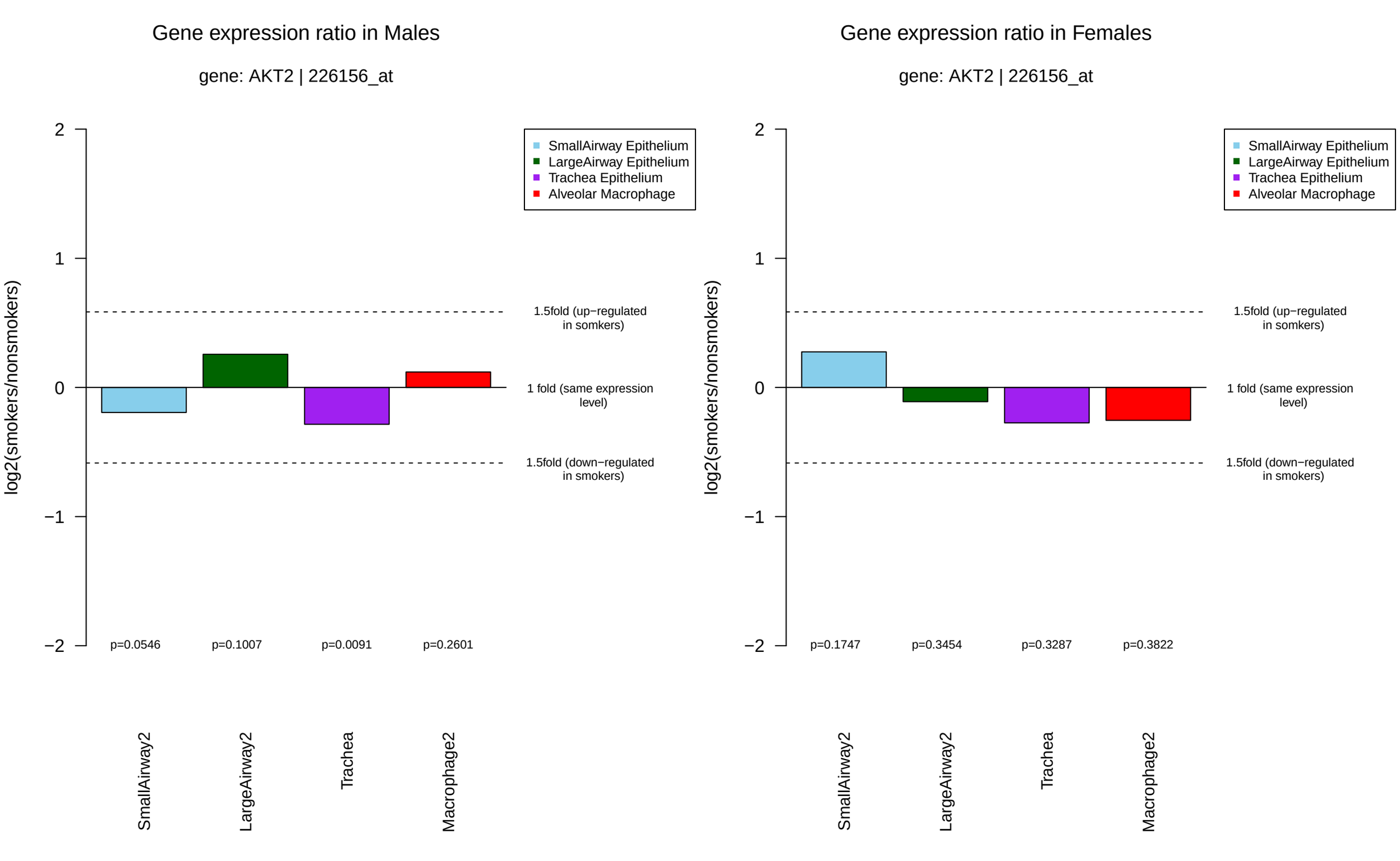
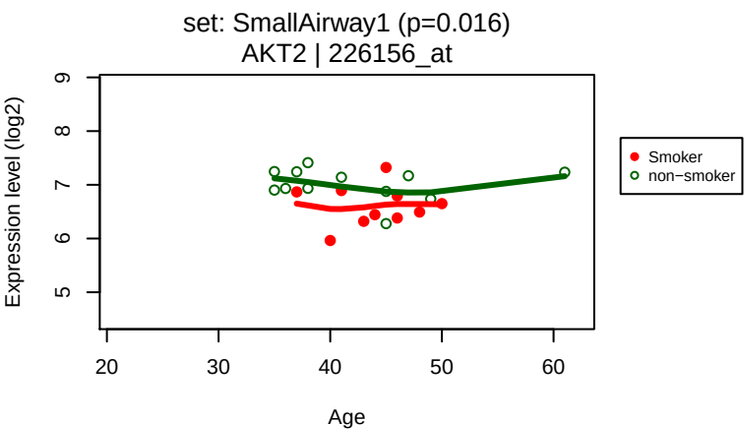
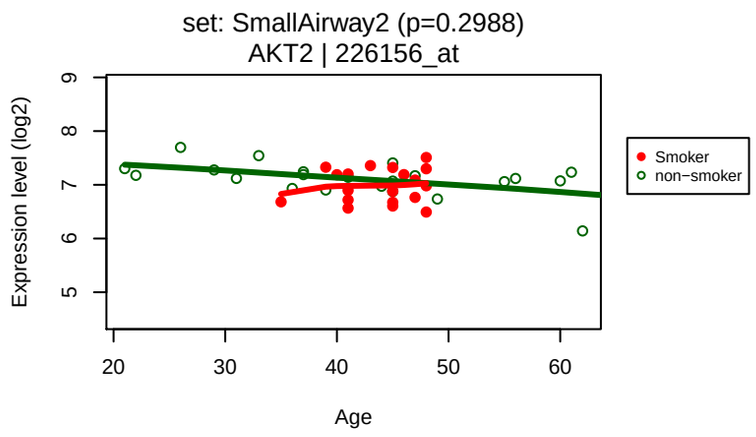
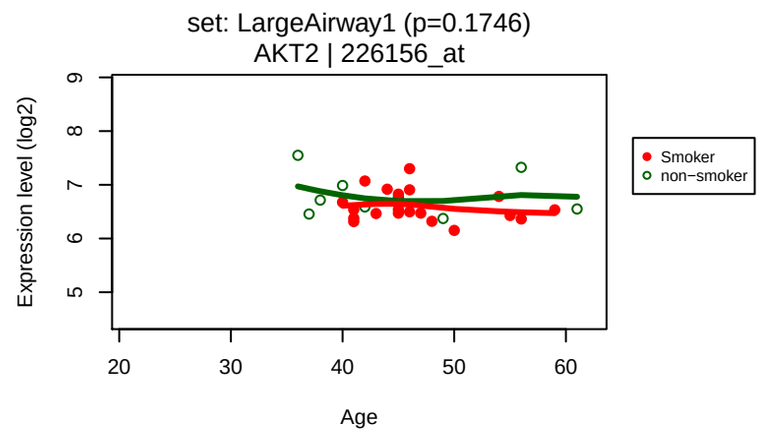

Gene expression of smokers and nonsmokers
gene: AKT2 | 226156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.016
p=0.2988
p=0.1846
p=0.1746
p=0.3556
p=0.0022
p=0.6359
p=0.9549
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

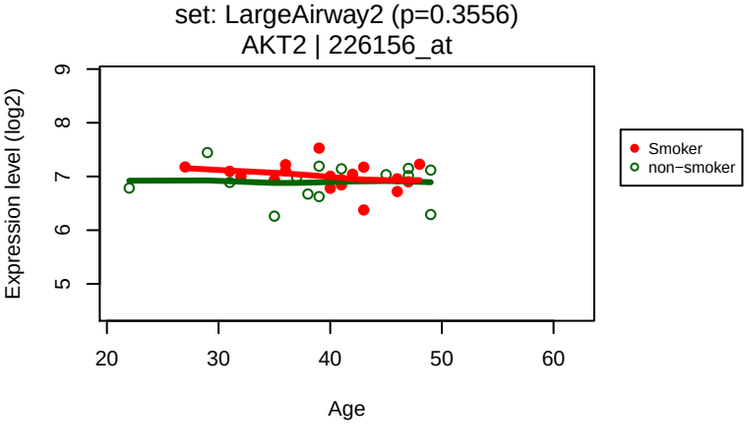

Gene expression ratio (smokers/nonsmokers)
gene: AKT2 | 226156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.016
p=0.2988
p=0.1846
p=0.1746
p=0.3556
p=0.0022
p=0.6359
p=0.9549
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

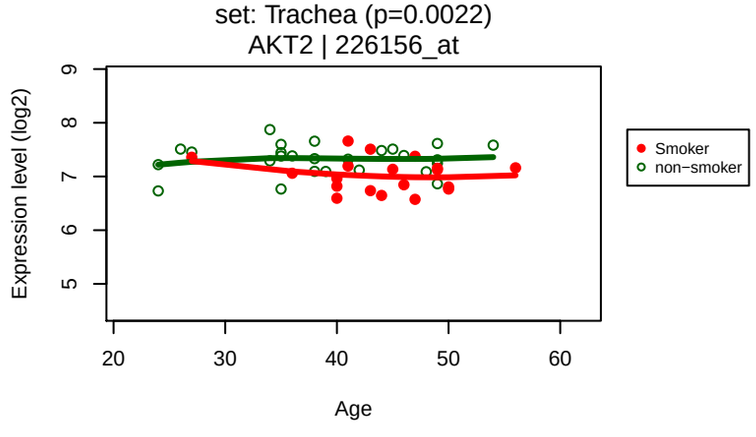

Gene expression ratio in Males
gene: AKT2 | 226156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0546
p=0.1007
p=0.0091
p=0.2601
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT2 | 226156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1747
p=0.3454
p=0.3287
p=0.3822
SmallAirway2
LargeAirway2
Trachea
Macrophage2

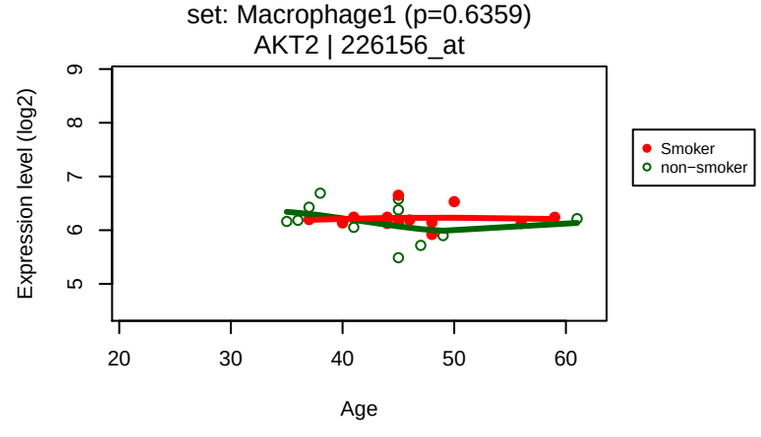

set: SmallAirway1 (p=0.016)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker

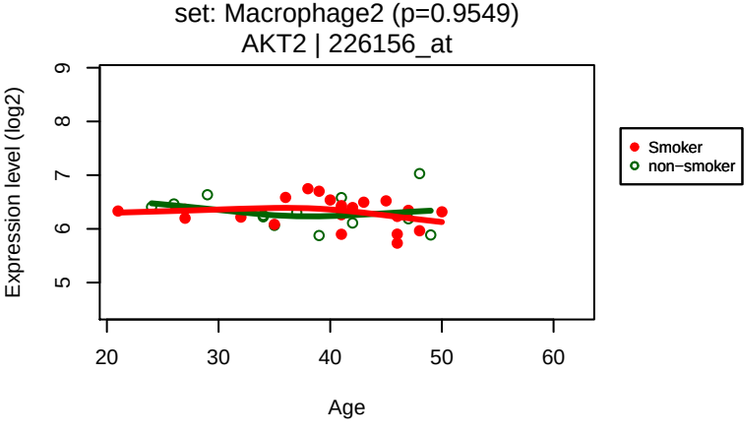

set: SmallAirway2 (p=0.2988)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker

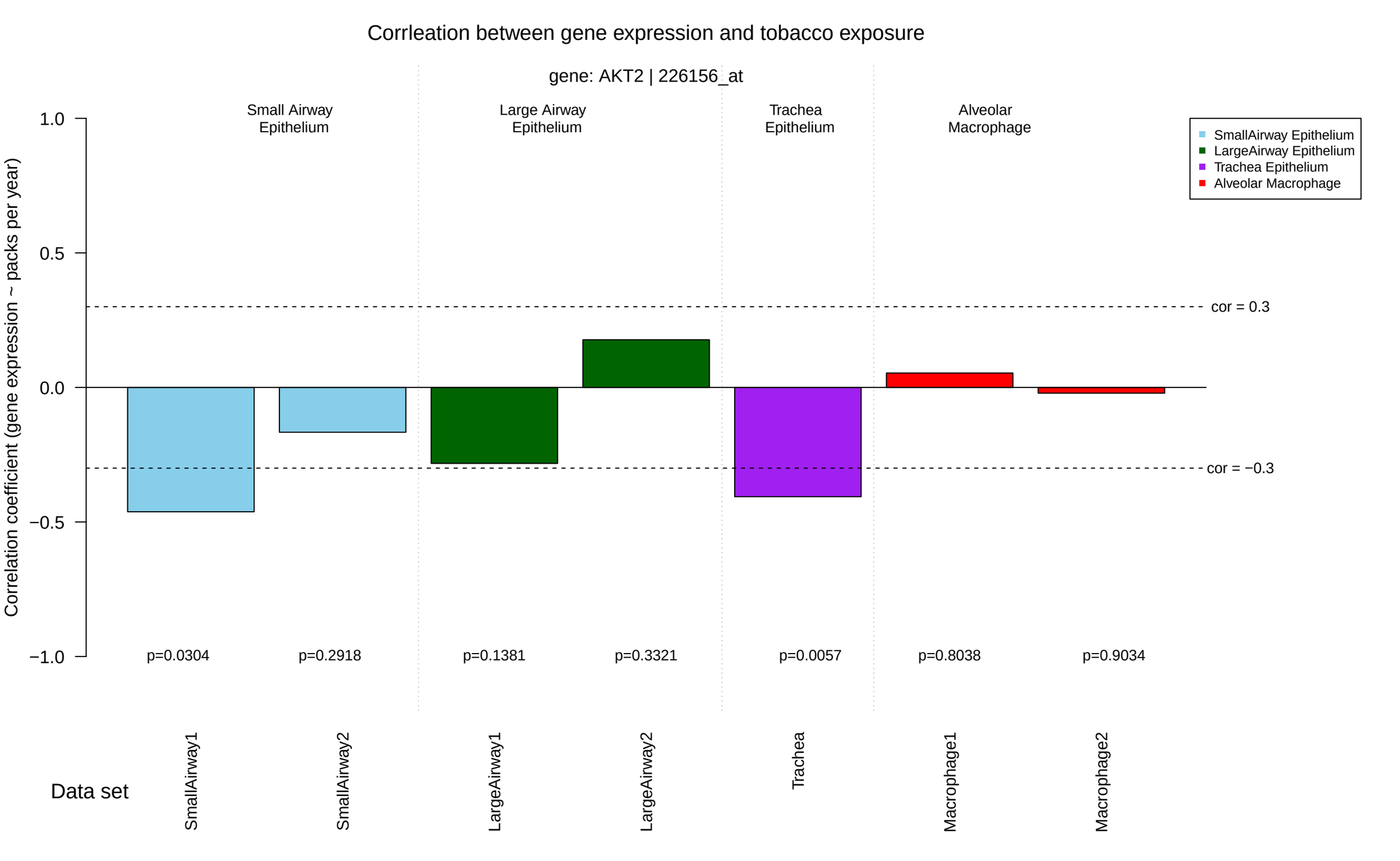

set: LargeAirway1 (p=0.1746)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3556)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6359)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9549)
AKT2 | 226156_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT2 | 226156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0304
p=0.2918
p=0.1381
p=0.3321
p=0.0057
p=0.8038
p=0.9034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2